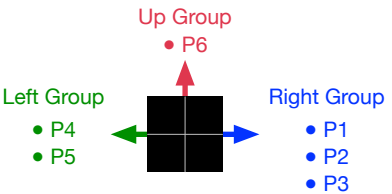
1. Group paths

ordered by direction cost (ascending)

grouped by primary direction

PathID	Direction Preference			
• P1	right (0.1)	up (0.6)	down (1.3)	left (1.8)
• P2	right (0.3)	down (0.7)	up (0.9)	left (1.6)
• P3	right (0.4)	up (0.8)	down (1.1)	left (1.9)
• P4	left (0.1)	down (0.5)	up (1.2)	right (2.0)
• P5	left (0.2)	up (0.3)	down (1.5)	right (1.8)
• P6	up (0.1)	right (0.5)	left (1.2)	down (1.7)



2. Order groups & paths

1. Right
1) P1
2) P2
3) P3

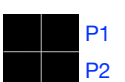
2. Left
1) P4
2) P5

3. Up
1) P6

3. Assign by (1st, 2nd) direction

assign P1
(right, up)

assign P2
(right, down)



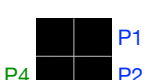
overflow of P3

4. Handle overflow

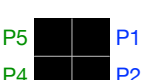
3. Up
1) P3
2) P6

P3 is assigned to up group, cost set to zero

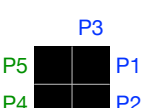
5. Repeat until complete



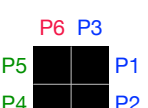
assign P4
(left, down)



assign P5
(left, up)



assign P3
(up, right)



assign P6
(up, right)